

ATHLETE DATA COMPANY

Athlete Intelligence Development & Data Metrics Blueprint

A product, data and scoring framework for capturing athlete, brand and ADC intelligence - and converting it into sellable commercial insights.

Cricket-first architecture | Expandable across sports | Version 1.0

Core thesis: ADC should not sell raw athlete data. It should sell better commercial decisions around athletes.

1. Executive Summary

ADC's long-term opportunity is to create a persistent athlete intelligence layer that connects identity, performance, audience demand, commercial value, brand fit and cross-event outcomes. The product should start with one sport (cricket), build a reliable longitudinal dataset, validate scoring logic, and only then expand to additional sports and event partnerships.

The recommended product architecture contains five intelligence engines: Performance Intelligence, Audience/Social Intelligence, Commercial Value Intelligence, Brand Matching Intelligence, and Growth/Potential Intelligence. Each score should be explainable, normalized, versioned and traceable to source data.

The key commercial loop is: Athlete Data -> Scoring -> Brand Decision -> Campaign/Event Outcome -> ROI -> Learning -> Better Scoring. This feedback loop is the core data moat.

Important principle: AI must not invent ROI. Start with deterministic calculations and validated data. Use ML/AI for normalization, classification, prediction, anomaly detection, entity resolution and explanations after enough historical data exists.

Recommended execution sequence

Select 1 Sport -> Build Athlete Dataset -> Define Athlete Tiers -> Build Performance & Audience Scores -> Track Across Events -> Capture Commercial Cost -> Measure Commercial Value & ROI -> Identify Undervalued/High-Growth Athletes -> Generate Sellable Insights -> Validate with Brands/Agencies -> Paid Pilots -> Event Partnerships -> Cross-Event Intelligence -> Scale Across Sports.

2. ADC Intelligence Architecture

ADC should separate raw facts from derived metrics and derived metrics from commercial predictions. This prevents opaque scoring and makes the system auditable.

Layer	Purpose	Examples
1. Source Data	Raw evidence from first-party, partner, public/licensed and athlete-submitted sources	Scorecards, athlete profiles, SportsTrade transactions, social metrics, campaign data, event data
2. Identity & Normalization	Create one persistent athlete identity and standardize data	Athlete ID, team history, event IDs, role mapping, metric normalization, duplicate resolution
3. Feature Store	Reusable calculated metrics used by scoring engines	Recent-form index, follower growth, engagement rate, sponsorship cost, audience geography
4. Intelligence Engines	Generate explainable scores	Performance, Audience, Commercial Value, Brand Match, Growth Potential
5. Insight Layer	Turn scores into decisions	Undervalued athletes, breakout candidates, sponsorship efficiency, cross-event comparisons
6. Activation Layer	Enable commercial action	Brand discovery, shortlist, outreach, campaign tracking, event reports, API/data products

Core entities

Entity	Minimum relationship model
Athlete	Athlete -> Sport -> Role -> Team -> Event -> Match -> Performance -> Audience -> Commercial outcomes
Brand	Brand -> Category -> Campaign -> Target audience -> Budget -> Athlete shortlist -> Partnership -> Outcome
Event	Event -> Competition level -> Athletes -> Matches -> Sponsors -> Audience -> Commercial outcomes
ADC	Data source -> Feature -> Score version -> Insight -> Recommendation -> Outcome -> model feedback

3. Athlete Data Model - What ADC Must Capture

Athlete data should be divided into identity, sporting context, performance, audience, commercial, verification and longitudinal outcome data. Do not store only the current state; retain history so ADC can calculate trajectories.

3.1 Athlete Identity & Profile

Category	Metadata Points	Why it matters
Identity	ADC Athlete ID, full name, DOB/age, gender, nationality, state/city, contact/agent (permissioned)	Persistent digital identity and deduplication
Sporting Profile	Sport, primary role, secondary role, batting/bowling style for cricket, dominant hand/foot where relevant	Role-aware comparisons
Career Context	Current team, previous teams, academy, federation, league level, years active	Career trajectory
Eligibility	Age category, professional/amateur status, competition eligibility	Discovery and compliance
Verification	Identity verified, team/event verified, achievement verified, social account verified	Trust score
Availability	Injury/availability status where lawfully sourced, active/inactive, selection status	Performance context

3.2 Cricket Performance Metrics

Role	Basic metrics	Advanced / contextual features
Batter	Matches, innings, runs, balls, average, strike rate, 4s, 6s, 50s, 100s, not-outs	Recent form, consistency, boundary %, dot-ball %, chase performance, pressure innings, opponent strength, venue/format split
Bowler	Overs, balls, runs conceded, wickets, economy, average, strike rate, maidens, dot balls	Powerplay wickets, middle/death economy, boundary conceded %, pressure wickets, opponent quality, venue/format split
Wicketkeeper	Catches, stumpings, run-outs, dismissals/match	Opportunity-adjusted dismissals, missed chances where available, batting contribution
All-rounder	Batting + bowling + fielding metrics	Role balance, multi-dimensional match impact, contribution consistency
Fielder	Catches, run-outs, direct hits, drops where available	Fielding opportunities, runs saved, boundary saves

3.3 Context Metrics - mandatory for fair scoring

Competition strength	International, franchise, first-class/domestic, state, district, academy/grassroots
Opponent strength	Team rating, player quality, ranking
Format	Test/First Class, ODI/List A, T20, other
Recency	Last 5 matches, 10 matches, 90/180/365 days
Opportunity	Innings, balls faced, overs, minutes, starts
Match situation	Chase, knockout, pressure phase, powerplay/death
Venue/environment	Home/away, venue profile, conditions where available

4. Audience / Social Intelligence Engine

This engine should measure attention, engagement, growth and fan conviction, not simply follower count. SportsTrade first-party activity can become a differentiated signal because it reflects fan intent rather than passive social following.

Signal Family	Methods to Capture	Derived Features
SportsTrade	Buy volume, sell volume, unique traders, transaction frequency, watchlists, holdings, holding duration, price/interest movement	Demand index, conviction index, demand velocity, buy/sell ratio, fan concentration
Social accounts	Followers/subscribers by platform, posts, impressions/reach if available, likes, comments, shares, saves, video views	Engagement rate, reach efficiency, follower velocity, content consistency
Buzz	Athlete mentions, hashtag volume, search interest, news coverage	Buzz index, share of voice, buzz velocity
Sentiment	Positive/neutral/negative mentions; topic clusters	Net sentiment, reputation risk, sentiment momentum
ADC first-party	Profile views, follows, searches, shortlist appearances, brand profile views	Discovery velocity, brand interest index
Audience composition	Geography, age bands, gender mix, language, interests where lawfully available	Brand audience-fit features

Suggested Social/Audience Score v1

Audience Score (0-100) = 30% Engagement Quality + 25% Growth Velocity + 20% SportsTrade Fan Conviction + 15% Reach/Buzz + 10% Sentiment/Reputation.

Weights are starting hypotheses. ADC should back-test them against actual campaign outcomes rather than treating them as permanent truth.

5. Brand Data Model - What ADC Must Capture

A Brand Match Score cannot be credible unless ADC understands the campaign objective. A sportswear launch, regional retail campaign and financial-services awareness campaign should produce different athlete rankings.

Category	Fields / Attributes
Brand Profile	Brand ID, brand name, industry/category, sub-category, geography, company size, target markets
Campaign Objective	Awareness, reach, engagement, acquisition, sales, launch, community, CSR, regional activation
Target Audience	Age, gender, geography, language, interests, income proxy if appropriate, sport affinity
Athlete Requirements	Sport, role, age band, gender, geography, competition level, performance threshold, social threshold
Commercial Constraints	Total budget, athlete budget, expected fee range, campaign duration, deliverables, exclusivity
Channel Preference	Instagram, YouTube, offline event, digital content, appearance, jersey/product integration, other
Brand Safety	Category conflicts, prohibited categories, reputation thresholds, exclusivity conflicts
Success KPI	Reach, impressions, engagement, clicks, leads, sales, app installs, footfall, media value, brand lift
Historical Outcomes	Athletes used, cost, campaign KPI, commercial value, ROI, renewal, qualitative feedback

Brand-side data is especially valuable because it creates the labels needed to train future recommendation models: which athlete was selected, what was paid, what happened, and whether the brand renewed.

6. Commercial Value Intelligence Engine

This engine answers: What commercial value did an athlete create relative to the investment? It should distinguish measured value from estimated equivalent media value.

Investment Category	Inputs	Outputs
Partnership Cost	Athlete fee, production cost, travel/activation cost, agency cost where relevant	Total campaign investment
Exposure	Incremental impressions, reach, broadcast/digital exposure, brand-visible content	Exposure value / equivalent media value
Engagement	Qualified likes, comments, shares, saves, views, interactions	Engagement value and efficiency
Traffic	Profile visits, clicks, landing-page sessions, QR scans	Traffic value
Conversion	Leads, installs, purchases, registrations, coupon use	Attributed conversion value
Content Asset	Brand-owned usable content, athlete-generated content, licensing period	Content replacement/equivalent value
Brand Lift	Awareness, consideration, recall or sentiment study where available	Incremental brand value indicator
Retention	Repeat campaign, renewal, long-term partnership	Commercial confidence signal

Core formulas

Commercial Value = Attributable Revenue/Contribution + Validated Media/Exposure Value + Engagement/Traffic Value + Other Measured Campaign Value.

Return Multiple = Commercial Value / Total Investment.

ROI % = (Commercial Value - Total Investment) / Total Investment x 100.

Example: if Athlete C costs INR 3 lakh and generates INR 21.3 lakh of validated commercial value, Return Multiple = 7.1x and ROI = 610%. The 7.1x should be called a return multiple, not ROI.

7. Brand Matching Intelligence Engine

Brand Match should be campaign-specific. It should rank athletes for a particular brief rather than assign one universal 'brand score'.

Brand Match Factor	Brand Match Criteria	Expected Starting Weight
Audience Fit	Target geography, age, interests, language, sport affinity	25%
Commercial Efficiency	Historical cost vs value, engagement efficiency, expected return	20%
Performance/Relevance	Performance score, role, competition level, current form	15%
Social Momentum	Audience score, follower/buzz growth, engagement quality	15%
Brand/Category Fit	Category affinity, lifestyle alignment, previous categories	10%
Budget Fit	Expected athlete fee vs campaign budget	10%
Risk & Availability	Exclusivity, reputation, availability, verification	5%

Brand Match Score = weighted fit score after mandatory filters and risk exclusions.

Example output: Athlete C - Match 87/100 - expected return range 5.8-7.0x - high growth - strong Gen-Z/regional fit - within budget. Predictions should include confidence intervals and explanation of the top drivers.

8. Growth / Potential Intelligence Engine

This is the engine that can eventually identify athletes before sponsorship prices catch up with attention and performance.

Inputs/Context	Signals
Performance trajectory	Recent score trend, competition progression, consistency, improvement rate
Selection trajectory	Academy -> state -> domestic -> franchise -> national progression
Audience trajectory	Follower growth, engagement growth, SportsTrade demand velocity, brand searches
Age/experience context	Age relative to competition level; experience curve
Opportunity	Upcoming tournaments, team selection, expected playing opportunity
Commercial gap	Current sponsorship cost versus peer commercial value
Risk	Injury/availability where lawfully sourced, inconsistent participation, reputation volatility

Potential output: Breakout Probability, Growth Potential Score, Undervaluation Index and Expected Commercial Value Range.

A high-value ADC insight is not 'Athlete X is popular.' It is: 'Athlete X is currently priced below comparable athletes despite accelerating performance and audience demand.'

9. ADC Metrics - What the Platform Must Measure

ADC also needs internal metrics to prove that its intelligence is accurate, useful and commercially valuable.

Area	Metric
Data Coverage	Athletes onboarded, verified %, sports covered, events covered, matches captured, data completeness %
Data Quality	Freshness, source confidence, duplicate rate, missing rate, reconciliation errors, anomaly rate
Scoring Quality	Score stability, back-test accuracy, rank correlation with future outcomes, calibration error
Prediction Quality	MAE/RMSE for numeric predictions; precision/recall for breakout classification; confidence calibration
Marketplace Utility	Brand searches, filters used, athlete profile views, shortlists, contact requests
Conversion	Search -> shortlist -> outreach -> deal -> campaign completion -> renewal
Commercial	Brand ARR/MRR, insight report revenue, data/API revenue, event partnership revenue, take rate if applicable
Brand Outcome	Median return multiple, campaign ROI, cost per engagement, cost per qualified reach, renewal rate
Athlete Outcome	Discovery growth, brand enquiries, sponsorships secured, commercial value growth
Event Outcome	Athlete profiles created, sponsor matches, data completeness, sponsor reporting value, renewal

10. Sellable ADC Insight Products

The monetizable product is the decision or benchmark produced from the data - not the raw dataset alone.

	Primary Audience	Primary Use Case
Athlete Intelligence Profile	Brands, agencies, teams	Evaluate one athlete before engagement
Brand-Athlete Shortlist	Brands/agencies	Choose athletes for a campaign
Undervalued Athlete Report	Brands/agencies/franchises	Find high-upside talent before price increases
Event ROI Report	Brands/event organizers	Measure athlete/sponsor efficiency within a tournament
Cross-Event Athlete Benchmark	Brands, teams, federations	Compare athlete performance and commercial trajectory across competitions
Emerging Talent Index	Brands, scouts, federations	Identify breakout athletes
Portfolio Optimization	Large sponsors/agencies	Allocate sponsorship budget across multiple athletes
Event Athlete Intelligence Dashboard	Event organizers	Athlete identity, performance, audience and sponsor insights
API / Data Feed	Platforms, federations, media	Embed verified athlete intelligence into external products

Potential commercial models

Brand SaaS subscription + premium insight reports + event/federation data partnerships + API/data licensing + campaign analytics. Avoid depending on one monetization model until willingness-to-pay is validated.

11. Event ROI Study - End-to-End Logic

A credible event study needs a pre-event baseline, in-event measurement and post-event commercial outcome window.

Stage	Capture	ADC output
T-30 / Pre-event	Athlete cost, current performance, followers, engagement baseline, buzz, brand deals, SportsTrade demand	Baseline and expected-value estimate
During event	Match performance, exposure, mentions, social activity, fan transactions, brand-visible content	Live momentum and event impact
T+7 / T+30	Follower delta, engagement delta, media value, traffic, conversions, sponsorship enquiries	Incremental commercial outcome
Cross-event	Repeat same athlete across events	Trajectory, benchmark and persistent athlete value

Illustrative comparison

Athlete	Cost	Performance	Audience Growth	Commercial Value	Return Multiple	ROI
A	INR 50L	91	+8%	INR 70L	1.4x	40%
B	INR 8L	87	+46%	INR 38.4L	4.8x	380%
C	INR 3L	81	+82%	INR 21.3L	7.1x	610%

Athlete C creates less absolute commercial value than Athlete A, but $7.1 / 1.4 = 5.07x$ higher commercial efficiency per rupee invested. This is the kind of insight ADC can sell.

12. Data Sources, Governance & Prerequisites

Scoring quality will be constrained by data rights and data quality. ADC should create a source registry and provenance trail from day one.

Source	Examples	Considerations
First-party	Athlete onboarding, brand campaigns, SportsTrade activity, ADC product behavior	Consent, purpose limitation, access control, retention
Event/Partner	Official event feeds, organizer registration, federation/team feeds	Contracts, schema/versioning, SLAs, permitted reuse
Licensed API	Scorecards, statistics, social/listening data	License scope, rate limits, attribution, commercial-use rights
Public data	Official results/pages, public social metrics, news	Terms compliance, provenance, freshness, reconciliation
Derived data	Scores, rankings, predictions, embeddings/features	Versioning, explainability, confidence, model governance

Minimum technical prerequisites

Capability	Requirements
Persistent IDs	Unique athlete, event, match, team, brand and campaign IDs
Canonical schemas	Sport-specific metric dictionary plus common cross-sport ontology
Data pipelines	Batch + near-real-time ingestion, validation, deduplication and reconciliation
Historical warehouse/lakehouse	Immutable raw layer plus curated analytical tables
Feature store	Versioned reusable features for scoring and ML
Scoring service	Configurable weights, normalization, role-specific rules, score versioning
Model layer	ML/LLM services only where useful; model registry and evaluation
Explainability	Reason codes, feature contribution, source references and confidence
Admin governance	Manual review, corrections, overrides, verification and audit logs
Privacy/security	Consent, RBAC, encryption, retention/deletion, applicable privacy-law controls
Monitoring	Data drift, model drift, pipeline health, score anomalies and quality dashboards

13. AI / ML Development Maturity

ADC should not start by training a complex model. Build intelligence in stages so every score has business meaning and can be validated.

Stage	Model	Use Case
V0 - Deterministic	Rules, percentiles, normalized weighted metrics	MVP scores; easy to explain and debug
V1 - Statistical	Peer benchmarks, z-scores/percentiles, regression, time decay	Better normalization and expected-value models
V2 - Supervised ML	Gradient boosting/ranking/classification/regression	Predict campaign outcomes, breakout probability, expected ROI
V3 - Recommendation	Learning-to-rank / recommender models	Campaign-specific athlete ranking
V4 - LLM layer	Natural-language query, summarization, evidence-backed explanations	Generate sellable narratives and analyst copilot experiences

LLMs should explain and synthesize evidence, not silently replace deterministic calculations. All generated insights should be traceable to underlying features and source timestamps.

14. Scoring Governance Rules

Rule	Implementation
Normalize before weighting	Convert metrics with different scales into comparable 0-100 percentiles or standardized values
Compare like with like	Role, format, competition level, age/experience and opportunity must influence peer groups
Use recency decay	Recent matches should usually carry more weight than old performance
Handle missing data	Do not treat missing as zero; use confidence/coverage indicators
Prevent small-sample distortion	Require minimum matches/balls/overs/campaign observations or reduce confidence
Separate fact vs estimate	Label measured commercial outcomes separately from modeled equivalent values
Version every score	Store score_version, feature_version, model_version, timestamp and inputs
Explain every score	Return top positive/negative drivers and confidence
Back-test	Test whether historical scores predicted future performance/commercial outcomes
Human governance	Allow authorized review/override with reason and audit trail

Recommended score object

Score = value (0-100) + percentile + confidence + peer group + score version + timestamp + top drivers + source coverage.

15. Recommended Development Roadmap

Phase	Execution	Completion
0. Definition	Choose cricket; define ontology, roles, data rights, score hypotheses and target buyer	Metric dictionary and source map approved
1. Data MVP	Build 300-500 athlete longitudinal records across 2-3 competitions	Reliable identity + >80% completeness on core fields
2. Scoring MVP	Performance + Audience v0; athlete tiers; confidence and explanations	Scores reproducible and manually validated
3. Commercial Study	Capture/estimate partnership cost and measured event outcomes	First event ROI benchmark report
4. Brand Pilot	Brand brief -> shortlist -> track outcome	At least one paid pilot/LOI and feedback on ranking utility
5. Learning Loop	Add Commercial Value + Brand Match + Growth models	Outcome-linked scoring and back-testing
6. Event Partnership	Offer organizer athlete intelligence dashboard/report	Event/federation data partnership
7. Scale	More competitions, sports, APIs and cross-sport ontology	Repeatable data acquisition and paid intelligence revenue

Do not make 'number of AI models built' an execution KPI. The important milestones are data coverage, predictive usefulness, paid decisions influenced and repeat commercial outcomes.

16. ADC North-Star & KPI Tree

Suggested North Star: Number of athlete-brand decisions made using ADC intelligence that produce a measurable commercial outcome.

Supply	Verified athletes, athlete coverage, event coverage, data freshness
Intelligence	Score coverage, confidence, accuracy, prediction calibration
Demand	Active brands/agencies, searches, shortlists, reports/API usage
Transactions/Actions	Brand-athlete contacts, campaigns initiated, partnerships
Outcome	Commercial value generated, median ROI/return multiple, renewals
Moat	Historical athlete-event observations, campaigns with outcome labels, cross-event continuity

The most defensible dataset is not simply 'millions of athlete records'. It is athlete x event x performance x audience x commercial outcome accumulated over time.

17. Final Product Blueprint

ADC's platform should ultimately answer five questions for every athlete:

Question	Engine
How good is the athlete right now?	Performance Intelligence
How much genuine attention and fan conviction do they command?	Audience / Social Intelligence
How commercially efficient are they?	Commercial Value Intelligence
How well do they fit this specific brand/campaign?	Brand Matching Intelligence
How likely are they to become more valuable before the market reprices them?	Growth / Potential Intelligence

This leads to a simple commercial proposition:

ADC helps brands, agencies, teams and sports organizations identify, evaluate and commercially engage athletes using persistent performance, audience and cross-event intelligence.

Data is the input. Better athlete decisions are the product. Historical outcomes are the moat.

End of blueprint - Athlete Data Company | Product & Intelligence Framework